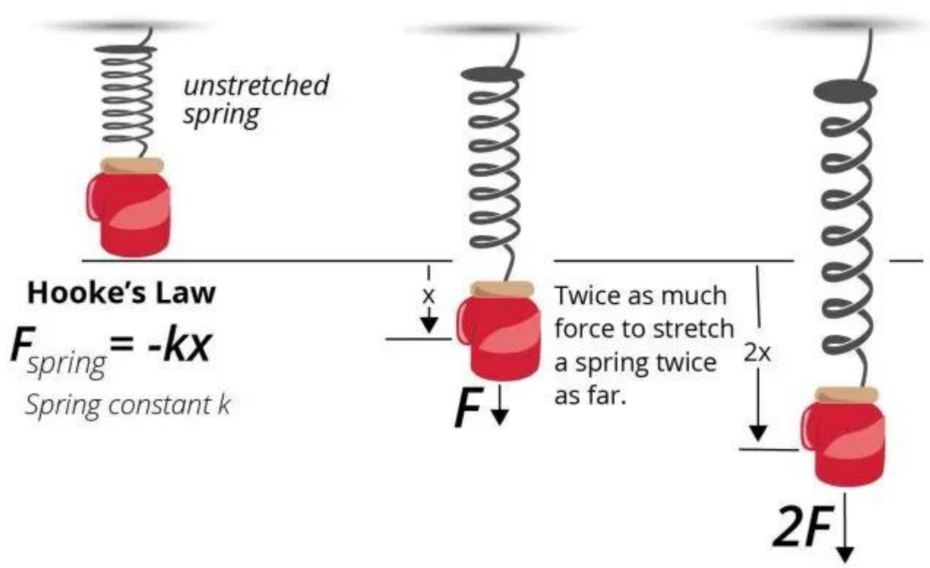
Graphical Analysis of the Hooke's Law

ENGS121 Mechanics Lab Section D

10 April, 2025

1. Introduction

The experiment is done to observe the pattern between the applied force to a spring and the amount it gets extended. In theory, springs obey Hooke's Law, which implies that the relationship between the extension of the spring and the force that acts on it must be linear. However, the measurements are done in an equilibrium where the force of the weight and the force of the spring are equal, hence the linear relation between those two can be hypothesized. Experiment is conducted to calculate and graphically represent the spring constant, and test the validness of theoretical formula $k \cdot l = k_1 \cdot l_1$.



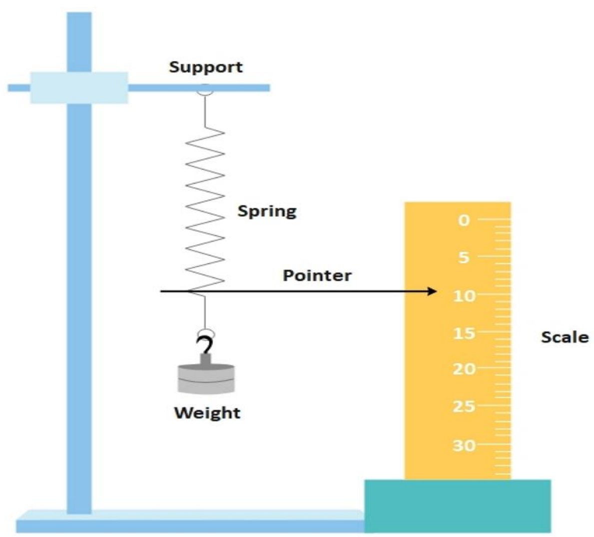
($F = -k\Delta x$, where k is constant, Δx is the amount the string gets extended.)

2. Measurements

Instruments

Name	Quantity	Properties
Ruler	1 pc	Wooden, 30.0 cm, resolution of 1 mm
Stand	1 pc	Stand
Thread	1 pc	Wool, 1 ft
Spring	1 pc	Lab Spring, with constant *k*
Weights	Set of Lab Weights	Different weights
Scales	1 pc	With an accuracy of 1 g

Experiment Procedure



The spring was mounted from the stand, and some parts of the string were looped with a thread. This loop will later prevent that part from further extension, from there onward. Different weights were attached to the spring and each time different lengths were recorded.

Length of the whole spring	6.7 cm
Length of the part without a loop	2.5 cm

Errors

Source of error	Type	Solution
Oscillations	Random	Can be prevented by stopping the string by hand and waiting for it to reach equilibrium

Variables

Variable	Value	Resolution
Attached Weight	Controlled	1 g
Length of Spring	Measured	1 mm
Oscillation	Controlled	1 mm

Spring Data Raw

Mass (g)	0	66	100	136	166	156	206	256	306	356
Length (cm)	6.7	10.1	11.25	12.7	14	13.6	15.5	18.4	19.8	21.2
Mass (g)	386	406	436	456	486	536	586	606	700	800
Length (cm)	22	22.5	23.3	23.4	24.2	25	25.9	26.8	28.1	28.5

3. Data Analysis and Plots

Spring Data SI Units Converted

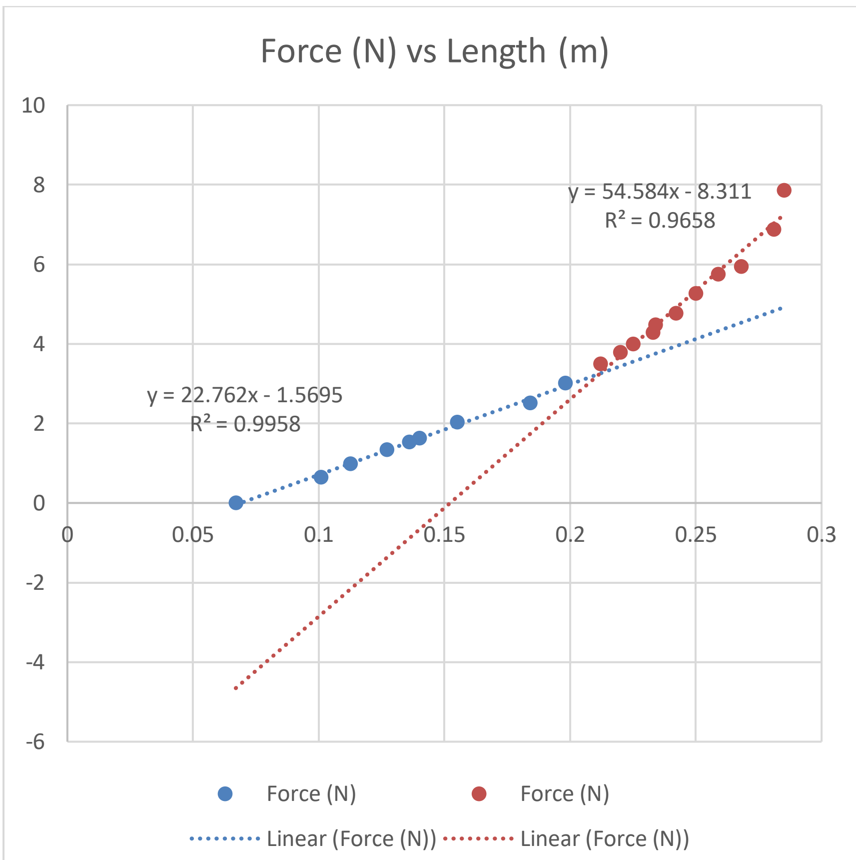
Mass (kg)	0	0.066	0.100	0.136	0.166	0.156	0.206	0.256	0.306	0.356
Length (m)	0.067	0.101	0.113	0.127	0.140	0.136	0.155	0.184	0.198	0.212
Mass (kg)	0.386	0.406	0.436	0.456	0.486	0.536	0.586	0.606	0.700	0.800
Length (m)	0.220	0.225	0.233	0.234	0.242	0.250	0.259	0.268	0.281	0.285

Grams and centimeters were turned to kilograms and meters accordingly.

Force – Length Data

Force (N)	0	0.647	0.981	1.334	1.628	1.530	2.021	2.511	3.002	3.492
Length (m)	0.067	0.101	0.113	0.127	0.140	0.136	0.155	0.184	0.198	0.212
Force (N)	3.787	3.983	4.277	4.473	4.768	5.258	5.749	5.945	6.867	7.848
Length (m)	0.220	0.225	0.233	0.234	0.242	0.250	0.259	0.268	0.281	0.285

At the equilibrium the magnitude of elasticity force and gravitational forces are equal, hence the formula $F = mg$ *can be used to calculate the magnitude of force.*



Data for Error Calculation

F (Predicted)	null	0.729	0.991	1.321	1.617	1.526	1.959
Length^2	0.00449	0.01020	0.01266	0.01613	0.01960	0.01850	0.02403
F (Predicted)	2.619	2.937	3.261	3.697	3.970	4.407	4.462
Length^2	0.03386	0.03920	0.04494	0.04840	0.05063	0.05429	0.05476
F (Predicted)	4.898	5.335	5.826	6.318	7.027	7.245	

Length^2	0.05856	0.06250	0.06708	0.07182	0.07896	0.08123	

The error in slope is

$$(\Delta k)^2 \approx \frac{1}{n \cdot \left(\overline{x^2} - (\bar{x})^2\right)} \cdot \frac{\Sigma\left(y_{predicted} - y_{experiment}\right)^2}{n-2}$$

The error in intercept is

$$(\Delta C)^2 \approx \overline{x^2} \cdot (\Delta k)^2$$

4. Conclusion and Evaluation

Using the formula $k \cdot l = k_1 \cdot l_1$

$k = 22.762\ N/m, l = 0.067$

$k_1 = 54.584\ N/m, l_1 = 0.025$

$k \cdot l = 1.525$

$k_1 \cdot l_1 = 1.365$

The difference between them is calculated to be ~12%

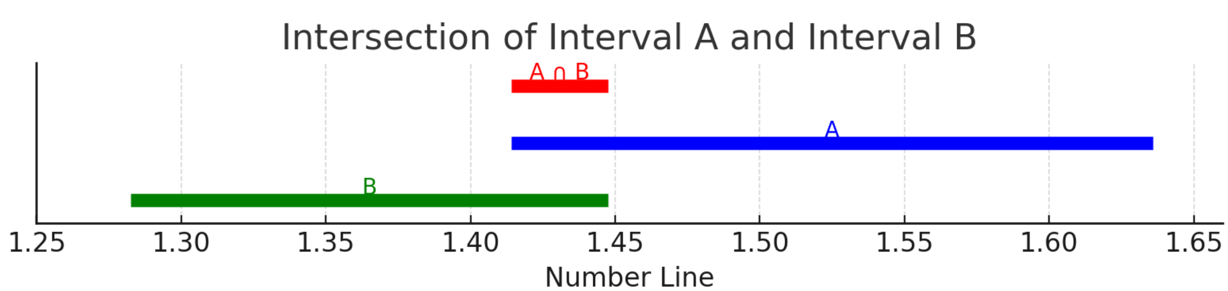
From Data analysis, we find that the error of the first slope $\Delta k = 0.75\ N/m$, for the second slope the error is $\Delta k_1 = 0.33\ N/m$.

Errors in intercepts for first and second line accordingly, $\Delta(k \cdot l) = 0.1109\ N$,

$\Delta(k_1 \cdot l_1) = 0.0825\ N$

Interval A = $k \cdot l\ \pm \Delta(k \cdot l)$ = [1.4141, 1.6359]

Interval B = $k_1 \cdot l_1 \pm \Delta(k_1 \cdot l_1) =$ [1.2825, 1.4475]



The breaking point of the graph corresponds to when the thread started to stop half of the spring from extending. From the graph it is visible that the spring constant of the entire spring is 22.8 N/m, and the spring constant of the part with no thread is 54.6 N/m. There is an interval of overlap in intersection of Intervals A and B, which means that the theoretical formula is valid in our experiment.

5. References

Hooke's Law. (n.d.). Digestible Notes. https://digestiblenotes.com/images/physics/alevel/hooke2.png